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RESEARCH ARTICLE

The birds and the trees: Avian ecosystem (dis)service perspectives and farmers' willingness to plant native trees in the agricultural landscape of the Galapagos Islands

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Abstract

1. Agricultural landscapes hold great potential for biodiversity conservation; however, this will require finding solutions that work for both people and nature. Increasingly, the conservation community is calling for more cross-disciplinary research integrating ecological questions with social and behavioural sciences for a more complete and successful approach to conservation.
2. Here, we used a mixed methods approach, including 53 in-person interviews, to examine how small-scale farmers in the Galapagos Islands perceive landbirds and their ecosystem (dis)services. We also studied farmers' motivations and hesitations in planting native trees on their farms, an action that was identified to be critical for landbird conservation in Galapagos in previous ecological studies.
3. We found that all native landbirds provided important cultural ecosystem services (CES) to farmers, with some variation between species. We also found that perceptions around pest control and pollination services were more variable and that landbirds were generally highly liked, even those who provided disservices such as crop damage.
4. Most farmers were willing to plant native and endemic trees on their farms, although typically only in small quantities. Farmers' main motivations for planting trees were instrumental (e.g. shade/freshness), while their primary hesitation was a perceived lack of space on their farms. The main predictor of willingness was the prior presence of native trees on their farm.
5. *Synthesis and applications*: Our results suggest tree-planting conservation efforts should be adaptive, context-dependent, and leverage incentives and community events for farmers. As a starting point, we recommend working with larger farms and with farmers interested in agrotourism for farmland restoration, recognizing that birdwatching has untapped potential in Galapagos. We further suggest

focusing conservation messaging for the local community on the CES we identify for specific Galapagos landbirds, since CES are linked to people's motivations to care for nature. Finally, more research is needed to explore what 'Galapagos identity' means since we found it to be associated with farmers' landbird perspectives and willingness to plant trees, and self-identity can drive pro-environmental behaviours. We also recommend more research to understand the pest control services landbirds may be providing, if any, in the agricultural zone.

KEYWORDS
agricultural restoration, biodiversity, conservation biology, conservation psychology, cultural ecosystem services, ecosystem services, farmer perceptions, neotropical birds

1 | INTRODUCTION

Agricultural expansion and intensification are some of the most important drivers of biodiversity loss (Kehoe et al., 2017). Between 1960 and 2019, we have changed about one-third of the total land surface of the Earth (Winkler et al., 2021). However, with proper management and the introduction of heterogeneity to farms, agricultural landscapes hold great potential for biodiversity conservation (Kremen & Geladi, 2024), especially as a complement to protected areas (DeFries et al., 2007; Kremen & Merenlender, 2018). Farms are largely privately owned, and thus, there has been a growing emphasis on private land conservation with an increasing need for cross-disciplinary work to integrate farmers' perspectives and knowledge with conservation practices (Ahnström et al., 2009; Drescher & Brenner, 2018). Here, we studied how small-scale farmers in the Galapagos perceive landbirds on their lands as well as their willingness, motivations, and hesitations to engage in conservation actions to support landbirds on their farms.

Considering local communities' perspectives and knowledge in conservation decisions is critical in their uptake, success, and longevity (Ahnström et al., 2009; Maas et al., 2021). Understanding human–nature relationships involves three key players: the landscape, the species that inhabit it, and the people (Figure 1). Traditionally, conservation biology is rooted in wildlife management, fisheries, and forestry and was focused on understanding the extent and complexities of biodiversity loss. Increasingly, the conservation community is calling for more cross-disciplinary research that goes beyond ecological inquiries, integrating social sciences to better understand people's perceptions, attitudes, and diverse values (Maas et al., 2021).

The ecosystem services and disservices framework can be especially helpful in understanding how wildlife, such as birds, affect people (Figure 1; Millenium Ecosystem Assessment, 2005). Birds, in particular, are known to provide important regulating services such as pest control, seed dispersal, and pollination (Sekercioglu et al., 2016). Conversely, they also provide 'disservices' such as damage to crops through direct consumption or indirectly through the consumption of natural enemies of crop pests (Pejchar et al., 2018). However, their contributions to cultural ecosystem services (CES)—defined as 'the nonmaterial benefits people obtain from ecosystems'

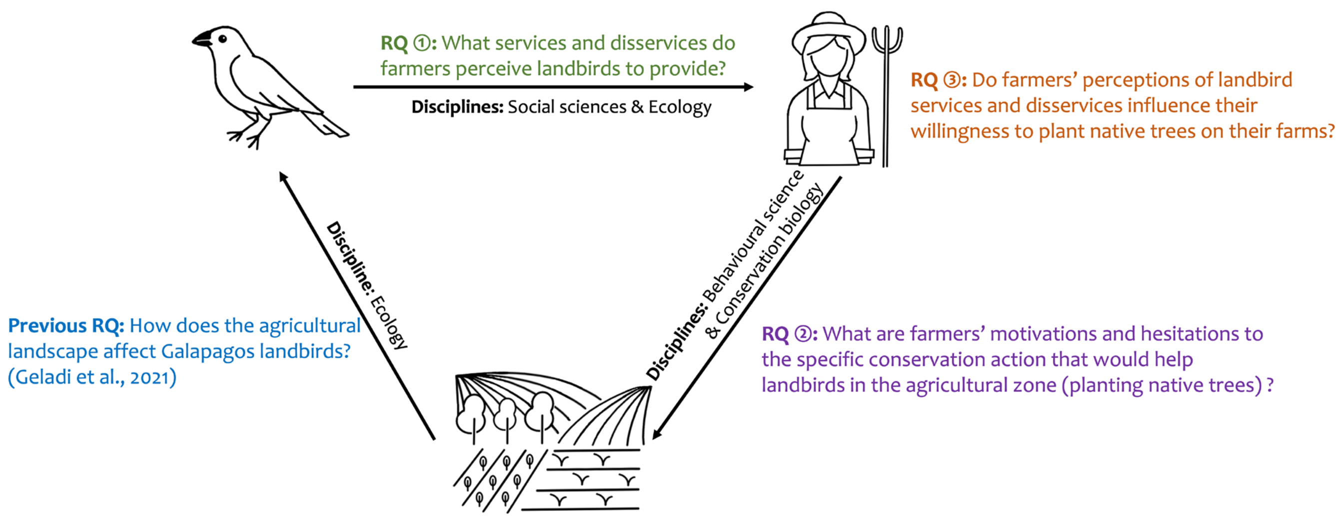

FIGURE 1 Conceptual framework representing our three key players: The agricultural landscape, Galapagos landbirds, and farmers, and the interactions between them that our research questions explore. Figure icons were illustrated by Julia Craig.

(Millenium Ecosystem Assessment, 2005), such as recreation, inspiration, aesthetic, and spiritual services—are less widely studied, particularly at the species level (Milcu et al., 2013; e.g. Echeverri et al., 2019). Nevertheless, there is great potential to capitalize on CES and integrate them into conservation plans as CES are important contributors to human well-being (Milcu et al., 2013).

Biodiversity conservation in agricultural landscapes requires considering human behaviour change with respect to land use and agricultural management (Figure 1) and can best be understood through behaviour science methods (Dayer et al., 2020; Nilsson et al., 2020). Recent studies have called for greater integration of social and human behavioural science with conservation (Balmford et al., 2021; Bennett et al., 2017), recognizing that behaviour change is key to driving both changes in personal actions and widespread patterns of behaviour change for conservation (Reddy et al., 2017). According to the theory of planned behaviour (Ajzen, 1985), attitudes and norms can predict human behaviour. Therefore, a major factor in human–wildlife interactions is the attitudes, norms, values, and previous experiences of individual humans in relation to the wildlife in question (Lischka et al., 2018; Manfredo & Dayer, 2004). However, in practice, conservation attitudes do not always translate to conservation behaviours (Nilsson et al., 2020), known as the 'attitude-behaviour gap' (Kollmuss & Agyeman, 2002). Most studies have focused exclusively on attitudes as a simple and measurable proxy for conservation behaviours despite the attitude–behaviour gap. Instead, human behavioural science recommends studies focus on the *specific* conservation action required to foster a deeper understanding of motivations and hesitations as a critical first step in designing effective biodiversity conservation interventions (Rare and the Behavioural Insights Team, 2019).

Therefore, instead of evaluating farmers' attitudes towards bird conservation, we focused on the specific behaviour that has been identified as directly contributing to landbird conservation in the agricultural zone of Galapagos: planting native trees. In the agricultural zone of Santa Cruz, where our study took place, previous research has shown that planting more trees, particularly native trees, could increase Galapagos landbird diversity and abundance (Geladi et al., 2021). This finding aligns with broader research showing that increased tree cover enhances bird species richness (Mellink et al., 2017; Mendenhall et al., 2016) and that even singular trees can increase bird species richness (Prevedello et al., 2018) through providing nesting, foraging, perching, singing, or roosting sites, or through providing stepping stones in agricultural landscapes (Pustkowiak et al., 2021). In the Neotropics in particular, tree richness and abundance at the local scale have been shown to have a significant impact on bird richness in coffee plantations (Gonzalez et al., 2024), with the positive effect of tree cover being especially important for specialist bird species (Valente et al., 2022), which are often more threatened.

The Galapagos archipelago is an area of unparalleled biological and conservation value due to its high levels of endemism, biodiversity, and its role, since Darwin, as a 'natural laboratory' for the study of evolution. However, human disturbances have increased drastically in the last 30 years, resulting in pronounced and continued declines in Galapagos landbird populations, particularly in the agricultural zone (Fessl et al., 2017). One of the primary drivers of landbird population decline is habitat loss (Geladi et al., 2021). On Santa Cruz Island, where our study is situated, the amount of natural vegetation in the agricultural zone has been reduced from 94% coverage in 1961 to 7% by 2018 (Alomía Herrera et al., 2022). The agricultural zone occupies what used to be the *Scalesia* zone, an endemic and structurally unique forest with important habitat for landbirds (Dvorak et al., 2012), as well as parts of the transition and *Miconia* zones that occur mostly on the southern slope of the island (Kelley et al., 2019). Currently, only an estimated 1.1% of *Scalesia* forest remains, all of which is found within National Park borders, as a result of direct human destruction for wood and clearing of forests for agriculture outside the park (Lundh, 2006; Mauchamp & Atkinson, 2010). Other contributing factors to Galapagos landbird declines include invasive plants (e.g. blackberry), which exacerbate habitat loss (Jäger et al., 2024), as well as the invasive deadly avian vampire fly (*Philornis downsi*), introduced vertebrate species, and diseases (Fessl et al., 2017).

In this study, we utilize the iconic landscape of the Galapagos Islands to examine the interrelationships between the agricultural landscape, landbirds, and farmers using an interdisciplinary research framework (Figure 1). Here, our first question asks: (1) What services and disservices do farmers perceive Galapagos landbirds to provide, and what factors are associated with these perceptions? Having identified planting native trees as the specific conservation action that can increase landbird diversity and abundance (Geladi et al., 2021), our second question asks: (2) What are farmers' motivations and hesitations to planting native trees on their farms, and what factors are associated with their willingness to plant native trees? Finally, we bring both questions together, asking: (3) Do farmers' perceptions of landbird services and disservices influence their willingness to plant native trees on their farms? Our findings provide broader takeaways for conservation actions for Galapagos landbirds in the agricultural zone.

2 | METHODS

2.1 | Study area

The research took place on Santa Cruz Island in the Galapagos Archipelago. Santa Cruz is the second largest island (986 km^2) and hosts the largest human population in the Galapagos Archipelago (~61% of Galapagos population; INEC, 2022). Consequently, it has the largest agricultural zone (114.4 km^2 or 11.6% of the island; Laso et al., 2020). The main zones that encompass the agricultural zone are: (1) small parts of the transition zone (80–200 m elevation); (2) the main area of the former *Scalesia* zone (200–550 m elevation); and (3) small parts of the *Miconia* zone (400–600 m elevation). The *Scalesia* zone is very humid and has the richest soils, hence the high rates of agricultural expansion in this zone (Kelley et al., 2019). The transition zone is composed of a mix of herbaceous plants as well as evergreen and deciduous trees representing a transition between the arid and more humid regions (See Section S1 for more information on the agricultural zone history and vegetation). We interviewed farmers from across the agricultural zone, which included the following townships (from east to west): El Cascajo, El Camote, Media Luna, Bellavista, Los Guayabillos, El Occidente, El Carmen, and Santa Rosa (Figure S1).

The Galapagos Archipelago is home to 34 native or endemic small landbirds, including four rails, two owls, and a hawk species. Of the endemic species or sub-species, meaning they are found only in Galapagos, 17 have been classified as threatened by extinction. Santa Cruz Island is home to 23 landbird species, which include nine Darwin's finches, a warbler, a mockingbird, two flycatchers, four rails, a dove, a martin, two owl species (Dvorak et al., 2012), a native cuckoo, and the introduced Smooth-billed Ani (*Crotophaga ani*) (Dvorak et al., 2012). Bird counts conducted in 1997, 2008, and 2019 documented a declining population of endemic landbirds on Santa Cruz, with the strongest declines occurring in the agricultural zone (Dvorak et al., 2012; Fessl et al., 2017; Geladi et al., 2021). In the agricultural zone, the most common landbirds are the Small Ground-Finch (*Geospiza fuliginosa*) and the Yellow Warbler (*Setophaga petechia aureola*), species that are also found in various other vegetation zones across the island (Geladi et al., 2021).

2.2 | Data collection

In June–July 2023, the interviewer (IG) conducted face-to-face structured interviews with a mix of open and close-ended questions with 53 farmers, each representing a distinct farm. We used face-to-face interviews because of highly variable access to technology and education levels among farmers, and because it allowed participants to ask clarifying questions (Santo et al., 2015). Although a complete list of farmers living on Santa Cruz was not available, there are a total of 530 official agricultural production units on Santa Cruz Island (MAG, 2022), meaning we sampled ~10% of production units.

Recruitment of interviewees primarily took place through the Ministry of Agriculture and Livestock (MAG). Our approach was convenience sampling, a non-random sampling method where interviewees were sampled mainly among farmers that already had a working relationship with the MAG or interviewer (Etikan, 2016). Convenience sampling was chosen for two practical reasons. First, it allowed us to identify willing participants, who already worked with MAG and helped build relationships of trust. Second, collaborating with MAG significantly improved our ability to reach and connect with farmers for face-to-face interviews. The MAG technicians' local expertise in navigating the agricultural zone streamlined the coordination of meetings and helped us overcome logistical challenges such as farmers' dispersed locations. Collaborating with a trusted organization such as MAG was considered a respectful approach. First, MAG colleagues compiled a list of farmers in Santa Cruz. Next, MAG technicians and the interviewer approached as many farmers as possible on the list. The technician introduced the farmer to the interviewer as an independent collaborator, and the interviewer then explained the purpose of the interview. If the farmer was willing, the interviewer returned at a later agreed-upon date to conduct the face-to-face structured interview with the farmer, without the presence of MAG personnel. Although most interviewees ($n=46$) were identified in this manner, other recruitment methods included the interviewer contacting farmers with whom they had pre-established contact ($n=5$), opportunistically approaching farmers ($n=2$), and sending out a recruitment poster to WhatsApp farmer association groups ($n=0$).

Interviews took on average 45–70 min each and were conducted in Spanish by IG to ensure consistency across interviewees. University ethics board approval was obtained (UBC BREB #: H22-00776). All interview participants agreed to participate and provided their informed oral consent, as written consent was deemed culturally inappropriate. The interviewer read aloud every question to the interviewee and recorded their responses on a paper copy, whilst the interviewee was handed a laminated version of the questions to follow along. For open-ended questions, the interviewer relied on written field notes recorded during the interview and permission from the farmer to record the interview. The recordings were used to verify the accuracy of the information collected through written field notes. A comparison between the field notes and recordings ($n=19$) resulted in a near perfect match.

Given our non-random sampling approach and limited sample size, our results may not be representative of all farmers on Santa Cruz Island. Therefore, when using the term 'farmers' in reference to our results, we specifically mean this sample of farmers. Nevertheless, data saturation for open-ended questions is often reached in under 25 interviews (Guest et al., 2006; Hennink & Kaiser, 2022), implying that the sample size for open-ended questions may adequately capture variability in response within our specific sample of farmers. For this reason, open-ended results may be interpreted with more confidence in our data.

2.3 | Interview instrument design

The structured interview incorporated a mix of open-ended and close-ended questions, with the latter presented as either ranked choice or Likert-scale items. We used a mixed methods approach, leading with the open-ended questions, which allowed more authentic, diverse, and specific responses from interviewees that could uncover deeper insights compared to close-ended questions. Close-ended questions allowed us to explore specific topics in a standardized manner, meaning we could ask every interviewee about precise items of interest they might not have considered without being prompted (e.g. pollination as a regulating service). We compare results from both open and close-ended questions to provide a comprehensive analysis (Creswell & Creswell, 2018).

The interview contained four sections (see Sections S13.4–S13.5 for the entire interview). The first section, 'Introduction', was about participants' farm and production. The second section, ‘Tree-planting’, asked questions about farmers' willingness, motivations, and hesitations for planting native and endemic trees on their farms. Endemic trees (estimated to be 43% of the floral taxa of the Galapagos Islands including *Scalesia* [*Scalesia pedunculata*] and Guayabillo [*Psidium galapageum*]) occur only in Galapagos, whereas native trees (i.e. Cat's claw [*Zanthoxylum fagara*]) are deemed to occur naturally there, but also occur elsewhere, such as on the continent of South America (Tye & Francisco-Ortega, 2011). We used the term ‘native’ to refer to both native and endemic tree species in the Galapagos since farmers did not distinguish between these terms. Motivation for planting native trees consisted of three value types (Himes et al., 2024): (1) intrinsic—the value of nature on its own, independent from people; (2) instrumental—values of nature based on what they can provide to people; and (3) relational—values of meaningful and often reciprocal human relationships with nature, often specific to a particular place, species, etc. (Chan et al., 2016; Pratson et al., 2023). Hesitations for planting native trees were coded into two categories: (1) biophysical—any concern relating to the biotic and abiotic natural world; and (2) personal—which referred to emotions, personal opinions, or circumstances.

The third section, ‘Birds’, first asked questions about farmers' general perspectives on landbirds, followed by detailed questions about three focal species, and less detailed questions about seven other landbirds. All landbirds included in the interview are commonly found in the agricultural zone of Santa Cruz (Geladi et al., 2021), with the exception of the Little Vermillion Flycatcher (*Pyrocephalus nanus*), which was historically found in the agricultural zone but is now at risk of extirpation on Santa Cruz (Anchundia et al., 2024). The three focal species were the Small Ground-Finch, the Yellow Warbler, and the Little Vermilion Flycatcher. These species are all endemic to Galapagos and were chosen to attempt to capture the range of perspectives that exist surrounding different landbird species in the agricultural zone (See Section S2, for further justification of species selection). Other birds included in less detail in the survey were: the Woodpecker Finch (*Camarhynchus pallidus*), the introduced Smooth-billed Ani (*C. ani*), the Galapagos Flycatcher (*Myiarchus magnirostris*), the Galapagos Mockingbird (*Mimus parvulus*), the Paint-billed Crake (*Mustelirallus erythrops*), the Galapagos Rail (*Laterallus spilonotus*), and the Dark-billed Cuckoo (*Coccyzus melacoryphus*). These include most remaining bird species in the agricultural zone that we thought farmers would be able to identify (note the focus on non-finch species since finch species can be difficult to distinguish; Geladi et al., 2022). Pictures of each species were shown to the interviewee (Sections S2 and S3) to determine whether they knew that species. If the farmer was not familiar with the species, we skipped the questions related to it. Finally, the fourth section included ‘Demographics’ questions.

Since the order of interview questions can influence participants' responses (Schuman & Presser, 1996), we placed the tree-planting section before the bird section to avoid priming participants to associate tree planting with bird conservation (Brekhus & Ignatow, 2019). Asking about tree-planting without linking it to birds aligns with behavioural science guidance to examine the specific conservation action (Rare and the Behavioural Insights Team, 2019), in this case native tree planting, independently of its intended outcome (i.e. bird conservation), in order to better understand baseline motivations. Where possible, questions were designed using feedback from semi-structured interviews conducted in 2019 (Geladi et al., 2022) and in consultation with MAG technicians who work day-to-day with Santa Cruz farmers. We pre-tested and refined the interview instrument with a MAG staff member and two local scientists from the Charles Darwin Foundation familiar with working with the farmer population in Santa Cruz.

2.4 | Participant characteristics

We primarily interviewed owners of farms ($n=44$), although some were family members ($n=3$), workers ($n=3$), or rented the land to manage the crops ($n=3$). The principal farm activity for each participant was: fruit and vegetable ($n=26$), pasture ($n=12$), coffee ($n=11$), poultry or pig ($n=3$), and agrotourism ($n=1$). Most farms (83%) also engaged in secondary activities. The size of the farms varied from 0.2 to 160 ha, with 60.4% of farms falling between 0.2 and 10 ha. Out of all participants, 79% relied on farm production as their main source of income and 83% spent 6–7 days of the week on their farm. Demographic characteristics of interviewees are summarized in Table 1. Cultural identity as ‘Galapagueños/as’ is an emerging construct

TABLE 1 Participant demographics.

Characteristics	Statistic ($n=53$)
Gender identity	
Male	45%
Female	55%
Age in years	
Mean (SD)	53.6 (13.2)
Range	32–83
Education level	
Primary school	45%
Secondary school	43%
Undergraduate degree	8%
Master's degree	4%
Born in Galapagos	
No	87%
Yes	13%
Years spent living in Galapagos	
Mean (SD)	35.9 (15.1)
Range	12–69
Galapagos identity	
‘I am not Galapagueño/a’	9%
‘I am a little Galapagueño/a’	21%
‘I am quite Galapagueño/a’	15%
‘I am Galapagueño/a’	55%

since the islands have only recently been colonized (20th century; Section S1), and many residents still have strong cultural ties to the mainland. However, cultural identity in Galapagos can be a contentious issue since there are tensions between the longer term residents and late comers, who are often viewed as outsiders (Chowa et al., 2023).

2.5 | Data analysis

The interviewer coded and analysed the interview data. Open-ended (qualitative) questions, such as 'Why do you like/dislike this bird?' were analysed descriptively using an inductive approach where we first identified codes, or short descriptions, from participants' responses. Then, where possible, these codes were grouped into broader themes (Braun & Clarke, 2006). For example, codes from the above question were categorized into higher order themes representing types of benefits, values, or negative effects associated with ecosystem services (Belaire et al., 2015; Chan, Satterfield, & Goldstein, 2012; Díaz et al., 2018; Echeverri et al., 2019; Millenium Ecosystem Assessment, 2005). Themes mentioned by fewer than 10% of interviewees were considered anecdotal (Teff-Seker et al., 2022). The classification of each response into its respective code and then theme was evaluated and verified by a second coder. In cases of disagreements, the coders engaged in a discussion to define the code in question clearly and reach a consensus on the appropriate classification. For open-ended questions that consisted of listing items, such as 'Which birds damage your crops?' we summed the number of times each response arose.

Close-ended (quantitative) questions, which include ranked choice and Likert-scale items, were analysed statistically by first examining the distribution of responses across the available categories. Then, to test whether responses significantly differed from one another, we used a non-parametric Friedman test for Likert-scale data followed by pairwise Wilcoxon signed-rank tests (if the Friedman test was significant) to determine which specific items differed. For ranked choice questions, we used a pairwise chi-squared test to test for differences in responses.

Finally, to explore associations between farmer perceptions of landbird (dis)services and farmers' willingness to plant native trees, we used various statistical approaches. First, we used the 'alpha' function in the 'psych' package to calculate Cronbach's alpha, using a cut-off of alpha >0.70, to measure internal consistency of pre-defined groupings for Likert-scale items (Revelle, 2023; Tavakol & Dennick, 2011). If alpha <0.7, we kept the Likert-scale items separate for downstream analysis. Next, we calculated the mean scores for the items in each grouping creating: (1) two constructs for general landbird perspectives (services and disservices); (2) four value constructs for motivations for planting native trees (relational, intrinsic, instrumental-biophysical, and instrumental-agrotourism); and (3) two constructs for hesitations about planting native trees (biophysical and personal). We used the general bird service and disservice constructs as dependent variables in models to test whether they were influenced by farm

TABLE 2 Multiple linear regression outputs for variables associated with general services and disservices perceptions of birds held by farmers.

Part A: Services

***Model*: General bird service perceptions~farm type + bird damage perceptions + pest control perceptions + age + Galapagos identity + education + frequency of religious events attendance + gender**

Term	Estimate	Std. error	*t* value	*p* value
Farm type 1: Poultry or pig	0.016	0.858	0.018	0.985
Farm type 2: Coffee	−0.239	0.798	−0.300	0.766
Farm type 3: Pasture	−0.038	0.793	−0.048	0.962
Farm type 4: Fruit and/or vegetable	0.010	0.774	0.013	0.990
Bird damage (*z*-scored)	−0.026	0.106	−0.247	0.806
Pest control (*z*-scored)	0.324	0.099	3.272	0.002**
Age (*z*-scored)	−0.114	0.127	−0.898	0.375
Galapagos identity (*z*-scored)	0.093	0.125	0.745	0.461
Education (*z*-scored)	0.046	0.115	0.398	0.693
Religion (*z*-scored)	0.151	0.106	1.425	0.162
Gender: Woman	−0.267	0.207	−1.287	0.206
Intercept	6.225	0.751	8.283	<0.001***
F (11, 39) = 1.808, p = 0.09				
R^2 = 0.338				
Adjusted R^2 = 0.151				

Part B: Disservices

***Model*: General bird disservice perceptions~farm type + bird damage perceptions + pest control perceptions + age + Galapagos identity + education + frequency of religious events attendance + gender**

Term	Estimate	Std. error	*t* value	*p* value
Farm type 1: Poultry or pig	1.346	1.198	1.123	0.268
Farm type 2: Coffee	1.277	1.115	1.146	0.259
Farm type 3: Pasture	0.888	1.108	0.801	0.428
Farm type 4: Fruit and/or vegetable	1.276	1.082	1.180	0.245
Bird damage (*z*-scored)	0.652	0.148	4.405	<0.001***
Pest control (*z*-scored)	0.149	0.138	1.075	0.289

(Continues)

TABLE 2 (Continued)

Part B: Disservices				
Model: General bird disservice perceptions~farm type + bird damage perceptions + pest control perceptions + age + Galapagos identity + education + frequency of religious events attendance + gender				
Term	Estimate	Std. error	*t* value	*p* value
Age (*z*-scored)	0.019	0.178	0.105	0.917
Galapagos identity (*z*-scored)	−0.387	0.174	−2.217	0.033*
Education (*z*-scored)	−0.419	0.161	−2.608	0.013*
Religion (*z*-scored)	0.092	0.148	0.621	0.538
Gender: Woman	−0.065	0.290	−0.226	0.822
Intercept	1.857	1.050	1.769	0.084
F (11, 39) = 3.644, p = 0.001				
R^2 = 0.507				
Adjusted R^2 = 0.368				

Note: *p*-value significance codes used are: ***$p<0.001$, **$p<0.01$, and *$p<0.05$.

type, bird damage perceptions, pest control perceptions, and demographic variables (Table 2). In a separate set of models, we used responses to the Likert-scale item, 'I would plant more native trees on my farm' as the dependent variable (Table 3) to examine: (1) whether the above-mentioned motivation and hesitation constructs, along with demographic variables, influenced willingness to plant native trees (Table 3a); and (2) whether farmers' perceptions of bird (dis)services (based on the constructs described above) influenced willingness to plant native trees, including the following covariates: existing presence or absence of native trees, amount of trees already present on the farm (Villamayor-Tomas et al., 2019), and demographic variables (Table 3b). We tested for collinearity between all independent variables for each model using Pearson's correlation for continuous variables, Spearman's correlation for ordinal variables and chi-squared tests for categorical variables. We removed any variables that had a correlation coefficient >0.5 (Dormann et al., 2013) or a significant association ($p<0.05$) in categorical tests. As a result, we excluded years on farm (correlated with age; $r=0.7$), total time in Galapagos (correlated with age; $r=0.8$), and worry about bird damage (correlated with perceived bird damage; $r=0.5$). Finally, we normalized the independent variables (where relevant) and performed a multiple linear regression analysis. We tested for collinearity in the regression using variance inflation factors (VIF) with the 'car' package and concluded that no variables were causing multicollinearity (all VIF <5) (Fox & Weisberg, 2019; O'Brien, 2007).

In the multiple linear regression models in which we explored predictors of farmers' willingness to plant native trees, we used the 'AICcmodavg' package (Mazerolle, 2023) to calculate the Akaike information criterion (AIC) to rank pre-defined candidate models which varied in the sets of covariates included in the model (Section S3). We then selected the model with the lowest AIC, verifying a positive *R*-squared, to select the best final model. Although statistical analyses were conducted for close-ended (quantitative) responses, we interpret the results with caution due to the relatively small sample size, which may limit the robustness and generalizability of the findings. All analyses were done in R (R Core Team, 2023).

3 | RESULTS

3.1 | Landbird services and disservices as perceived by farmers

3.1.1 | General perspectives held by farmers about landbird (dis)services

Farmers generally agreed that landbirds in the agricultural zone provided services ($\bar{x}=6.03$, SE = 1.16) and generally disagreed that landbirds provided disservices ($\bar{x}=2.99$, SE = 1.74). For services, the level of agreement on landbirds in the agricultural zone being useful in pest control was lower than the other ecosystem services except 'I think landbirds in the agricultural are owners of the land' ($\chi^2_{(5)}=45.4$, $p<0.001$; Figure 2; Table S1). For disservices, farmers agreed more with landbirds being an economic burden and landbirds being bothersome compared to landbirds being too noisy or unpleasant to look at, with which they disagreed ($\chi^2_{(3)}=76.4$, $p<0.001$; Figure 2; Table S1).

When asked whether landbirds damage crops on their farm, 'a little bit' ($n=21$) was the most common answer ($\chi^2_{(4)}=16.15$, $p=0.003$; Figure S2; Table S2). As a follow-up, out of farmers who perceived damage ($n=42$), most farmers were 'a little bit' worried about this damage by landbirds ($n=16$; $\chi^2_{(4)}=12.05$, $p=0.02$; Figure S2; Table S2). Finches were the most mentioned landbird to damage crops ($\chi^2_{(5)}=111.36$, $p<0.001$; $n=37$; Figure S2; Table S2), followed by the Common Gallinule (*Gallinula galeata*; $n=12$). Crops damaged by landbirds, according to farmers, included tomatoes, peppers, corn, and cucumbers (Figure S2).

When asked whether landbirds help with controlling pests on crops, 'a little bit' was the most common answer ($n=24$; Figure S3; Table S3). The Yellow Warbler was listed most often as a landbird that provides pest control, followed by finches, and the Smooth-billed Ani (Figure S3). Landbirds consume various pests, according to farmers, including caterpillars ($n=34$), butterflies ($n=20$), ticks, crickets, wasps, and mice (all $n<10$; Figure S3).

Finally, 55% of farmers brought up, unprompted, concerns about the decline of landbirds on Santa Cruz Island (Section S4).

3.1.2 | Exploring potential drivers of general perspectives held by farmers about landbird (dis) services

The main variable associated with general landbird services perspectives by farmers was their perception of how much landbirds contributed to pest control ($\beta=0.336$, $p=0.002$; Table 2a). The

TABLE 3 Multiple linear regression outputs for variables associated with farmers' willingness to plant native trees on their farms and exploration if farmers' perceptions of bird (dis)services influence willingness to plant native trees.

A. What might affect farmers' willingness to plant native trees on their farms?

***Model*: Willingness to plant native trees~Motivation$_{intrinsic}$ + Motivation$_{instrumental_biophysical}$ + Motivation$_{instrumental_agrotourism}$ + Motivation$_{Relational_Value}$ + Hesitation$_{biophysical}$ + Hesitation$_{personal}$ + Amount of trees**

Term	Estimate	Std. error	*T* value	*p* value
Motivation—Intrinsic value	0.362	0.317	1.143	0.263
Motivation—Instrumental-biophysical	0.521	0.169	3.078	0.005**
Motivation—Instrumental-agrotourism	0.260	0.116	2.234	0.034*
Motivation—Relational values	−0.929	0.373	−2.497	0.019*
Hesitation—Biophysical	−0.538	0.260	−2.074	0.048*
Hesitation—Personal	0.221	0.228	0.970	0.341
Amount of trees already on farm	0.003	0.008	0.459	0.650
Intercept	5.788	3.016	1.919	0.066
F (7, 26) = 4.178, p = 0.003				
R^2 = 0.529				
Adjusted R^2 = 0.403				

B. Do farmers' perceptions of bird (dis)services influence willingness to plant native trees?

***Model*: Willingness to plant native trees~general bird service perceptions + general bird disservices perceptions + native tree presence + amount of trees**

Term	Estimate	Std. error	*T* value	*p* value
General bird services average	0.564	0.368	1.532	0.134
General bird disservices average	−0.361	0.233	−1.550	0.130
Native tree presence	1.727	0.749	2.307	0.027*
Amount of trees already on farm	0.005	0.009	0.534	0.596
Intercept	1.190	2.862	0.416	0.680
F (4, 38) = 3.659, p = 0.01				
R^2 = 0.278				
Adjusted R^2 = 0.202				

Note: *p*-value significance codes used are: ***$p<0.001$, **$p<0.01$, and *$p<0.05$.

main variable associated with general landbird disservice perspectives by farmers was their perception of how much landbirds damaged crops ($\beta = 0.648$, $p < 0.001$; Table 2b). Other contributors to perspectives on landbird disservices were Galapagueño/a identity (farmers who identified more with being from Galapagos agreed less with negative Likert-scale items about landbirds [$\beta = -0.385$, $p = 0.031$]) and education (more educated farmers agreed less with negative Likert-scale items [$\beta = -0.415$, $p = 0.012$; Table 2b]).

3.1.3 | Species-specific landbird (dis)services

Small Ground-Finch

All farmers interviewed recognized the Small Ground-Finch, although recognition to the species level is likely variable (Section S4). Farmers had a range of perspectives about whether or not they liked this bird (Figure 3b; Table S5). Open-ended responses highlighted a wider range of traits farmers associated with CESs (compared to Likert-scale responses) provided by the Small Ground-Finch (Table 4). Farmers liked the Small Ground-Finch because it brings joy, for its aesthetics, for being native, and for its existence value (Table 4). The most common reason farmers reported not liking the Small Ground-Finch was that it harms crops (Figure 3a; Table 4). Likert-scale responses revealed a higher range of perspectives regarding ecosystem disservices and regulating ecosystem services (Figure 4; Table S4). Although many interviewees did agree that they didn't like the Small Ground-Finch because it harms crops, the Likert-scale responses reveal more of a division in opinion (Figure 4a; Table S6). Interviewees also agreed with the Likert-scale item 'I don't like this bird because it disperses invasive plants', even if it did not come up often in open-ended responses (Figure 4a; Table 4; Table S6). Finally, Likert-scale responses also revealed that there was less strong agreement among interviewees for liking Small Ground-Finches for providing regulating services (pest control and pollination) in comparison with cultural services ($\chi^2_{(8)} = 146$, $p < 0.001$; Figure 4a; Table S6).

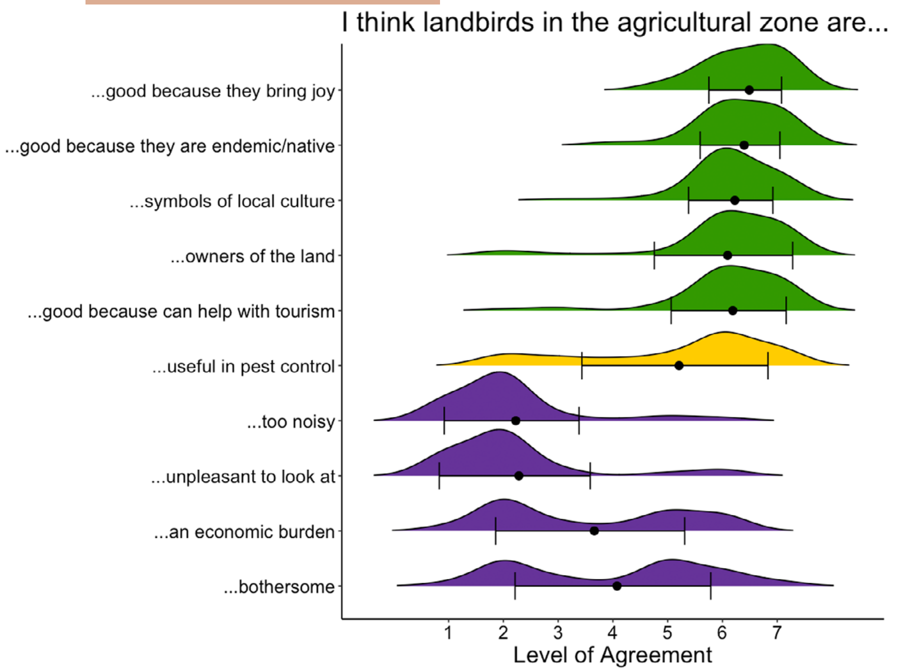

FIGURE 2 Distributions of Likert-scale responses to general landbird ecosystem service and disservice perceptions. Each dot represents the mean value of the level of agreement and the lines represent standard deviations from the mean. The level of agreement was assessed on a seven-point Likert scale: 1-strongly disagree, 2-disagree, 3-somewhat disagree, 4-neutral, 5-somewhat agree, 6-agree, 7-strongly agree.

Yellow Warbler

Almost all farmers interviewed (52/53) recognized the Yellow Warbler (Section S4). Most farmers interviewed liked the Yellow Warbler 'a lot' followed by 'quite a bit' ($\chi^2_{(2)} = 29.6$, $p < 0.001$; Figure 3d; Table S5). Based on open-ended responses, the main reason farmers liked the Yellow Warbler was because it was pretty and/or had a nice colour (Figure 3c; Table 4). Open-ended responses also revealed many farmers liked the Yellow Warbler for being harmless (Figure 3c; Table 4), meaning it does not damage their crops, which is a lack of disservice reflected in Likert-scale item responses (Figure 4b). Finally, interviewees also liked the Yellow Warbler for its behaviour, joy it brings, and companionship (Figure 3c; Table 4). Likert-scale responses reflected open-ended responses through agreement or strong agreement with nearly all CES items, except for 'This bird provides spiritual value' ($\chi^2_{(8)} = 130$, $p < 0.001$; Table S7). In comparison with the Small Ground-Finch, Likert-scale responses revealed less variation, especially among disservices (SE = 0.05 vs. 0.11, respectively; Figure 4; Table S4). More specifically, Likert-scale responses revealed that most farmers disagreed that Yellow Warblers contribute disservices, which was reflected in the lack of disservice comments in open-ended responses (Table 4). However, there were varying perspectives on interviewees not liking the Yellow Warbler because it disperses invasive plants (Figure 4b; Table S7). Finally, similar to the Small Ground-Finch, there was some variation in regulating service perceptions, and these differed from CES perceptions ($\chi^2_{(8)} = 130$, $p < 0.001$; Figure 4b; Table S7).

Little Vermilion Flycatcher

Most interviewees (51/53) recognized the Little Vermilion Flycatcher. However, most interviewees had either never seen this bird ($n = 18$) or had not seen it in over 10 years ($n = 23$; Section S4). Compared with the other focal species, interviewees liked this bird the most, with the 'I like it a lot' category scoring much higher than other categories ($\chi^2_{(2)} = 55.8$, $p < 0.001$; Figure 3f; Table S5). Open-ended responses revealed that the overwhelming reason for liking the Little Vermilion Flycatcher was that it was pretty and colourful (Figure 3e; Table 4), revealing these as the two most important traits associated with CES compared to Likert-scale responses, where interviewees strongly agreed with all CES Likert-scale items (Figure 4; Tables S4 and S8). In open-ended responses, interviewees also mentioned liking this bird because of its behaviour, something Likert-scale responses did not capture (Table 4). The only two Likert-scale items that had some diversity in perceptions were 'This bird provides spiritual value' ($\chi^2_{(5)} = 29.1$, $p < 0.001$) and 'I don't like this bird because it disperses invasive plants' ($\chi^2_{(1)} = 12.2$, $p < 0.001$; Figure 4c; Table S8).

Other species

Among the non-focal species, farmers liked the Galapagos Mockingbird, Galapagos Flycatcher, Galapagos Rail, and Dark-billed Cuckoo, albeit for different reasons as revealed by open-ended responses (Figures S4 and S5). Farmers had slightly more varying opinions about liking the Woodpecker Finch and Paint-billed Crake (*M. erythrops*), and most disliked the Smooth-billed Ani (hereafter: 'Anis'; Figure S4), which is an introduced landbird, often considered to be a damaging invasive (Cooke et al., 2019). Remarkably, farmers recognized a positive unique trait associated uniquely with each native landbird and a negative unique trait for the one introduced species (Figure S5). For example, participants liked the Galapagos Mockingbird for its recognizable songs and the Woodpecker Finch for its unique tool-using ability when feeding (Figure S5). The main reason farmers disliked Anis was because they had heard that they bother or eat other birds or their eggs. Despite this, some farmers still liked Anis a little bit, mainly because of pest control services (Figures S4 and S5). We did not include open-ended question results for the Paint-billed Crake as this species was often confounded with the Common Gallinule (Section S4).

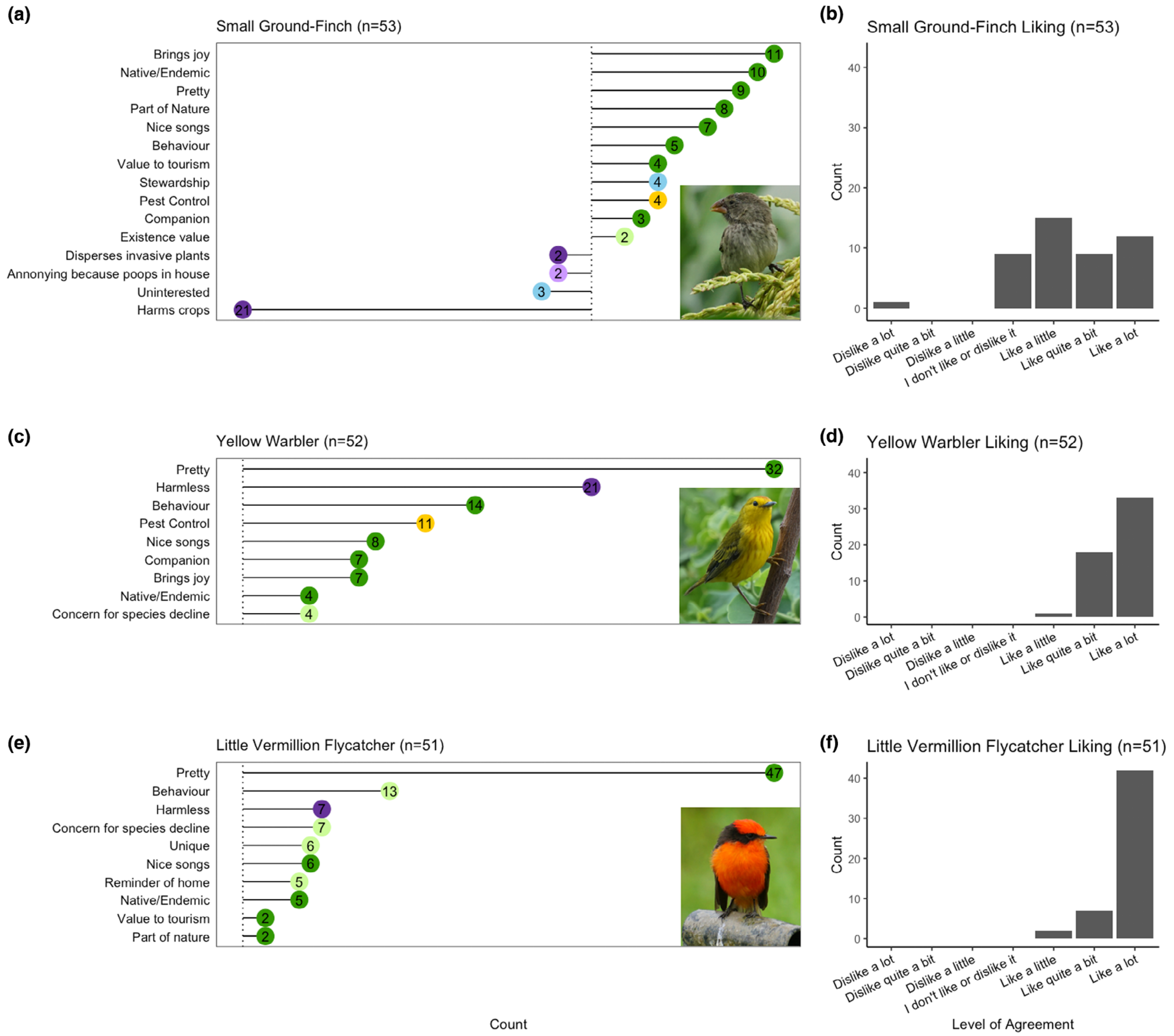

FIGURE 3 Species-specific perspectives of farmers for three focal Galapagos landbird species. Panels a, c and e show interview codes in response to the follow-up open-ended question: 'Why do you like or dislike this bird?'. Panels b, d and f show ranked choice responses to 'How much do you like or dislike this bird?'. Only codes with $n > 1$ are shown on the graph. Green colours represent codes (traits) associated with cultural ecosystem services, yellow represents codes (traits) associated with regulating ecosystem services, and purple colours represent codes (traits) associated with disservices (or lack thereof, see 'harmless'), and blue represents 'other' (see Table 4). Darker shades of the colour indicate this thematic code matched a Likert-scale item and lighter shades of colour indicate this thematic code did not match a Likert-scale item (see Table 4), adding more nuance to our understanding of species-specific perspectives. All pictures were taken by I. Geladi.

3.2 | Farmers' willingness, motivators, and hesitations to planting native trees on their farms

3.2.1 | Farmers' willingness to plant native trees on their farms

Although most farmers agreed that it is important to conserve nature in Galapagos through planting more native trees ($\bar{x} = 6.40$, SE = 0.09), there was some discrepancy in agreement as to whether they would be willing to plant more native trees on their own farms ($\bar{x} = 5.20$; SE = 0.24; Figure S6). Farmers were slightly more willing to help plant native trees on other farms ($\bar{x} = 5.89$, SE = 0.18; Figure S6). When asked how much of their property farmers were willing to plant trees (either native or non-native), 30 (out of 53) interviewees were willing to plant trees on 1%–25% of their land ($\chi^2_{(4)} = 48.5$, $p < 0.001$; Figure S6; Table S9) with 83% ($n = 25$) of interviewees preferring to plant native trees, 23% ($n = 7$) preferring to plant non-native trees, and 27% ($n = 8$) having no preference between planting native and non-native trees. We found that 90% of farmers already had at least one native tree on their property. Most of these trees (86% of

TABLE 4 Species-specific landbird ecosystem (dis)service perspectives of farmers.

Ecosystem service	Theme (type of benefit, value, or negative effect)	Code (trait)	*n*	Examples of responses	Likert statement	Likert score (mean + SE)
Small Ground-Finch						
Cultural	Activity[a,b]; Place[a,b]	Brings joy	11	"I like seeing the birds on my farm."; "They are joyful."; "Sometimes I call them and they come to see me in the morning."	I like this bird for their company	5.38 ± 0.22
		Behaviour	5	"They are tame."; "I like them because they are mischievous."; "I like them for the way they fly."		
		Companion	3	"Sometimes I call them and they come in the morning."; "They live with us."; "Before, they would come sleep in the house and I had 5–7 albino birds here."		
	Aesthetic[a,b,c]	Pretty	9	"It is nice to see them."; "They are pretty"; "They adorn nature."	I like this bird because it is pretty or/and has nice songs	5.58 ± 0.21
		Nice songs	7	"(I like them) for their song."; "In the morning they sing and bring joy."; "Their song draws my attention. When you hear birds, it is really nice to hear them every morning and it means you have other species nearby."		
		Colour	1	"(I like them) because they are black."		
	Cultural heritage[a]; Identity[a,c]	Native/Endemic	10	"They are birds from here."; "They are endemic."; "They are endemic so we are the intrusive ones."	I like this bird because it is endemic/native to Galapagos	6.04 ± 0.16
	Existence value[a]	Part of Nature	8	"They are part of nature."; "They are part of the ecosystem."; "They are abundant and live here."	I like this bird because it is part of the ecosystem	6.26 ± 0.14
		Existence value	2	"It would be sad if there were no more."; "They have the right to live."	–	–
		Evolutionary importance	1	"They are attractive because of evolution and their different shapes and sizes."	–	–
		Concern for species decline	1	"I suffer now that there are not many."	–	–
	Tourism[d]	Value to tourism	4	"I like them because they are an important part of tourism, they make up for the damage they do."; "They bring tourism."; "They are the attraction of Galapagos."	This bird is good because it can help for ecotourism	6.09 ± 0.15
	Bequest[a,c]	–	–	–	This bird is important for future generations	5.98 ± 0.18
	Spiritual value[a]	–	–	–	This bird provides spiritual value	5.19 ± 0.21

TABLE 4 (Continued)

Ecosystem service	Theme (type of benefit, value, or negative effect)	Code (trait)		*n*		Examples of responses	Likert statement	Likert score (mean + SE)
Regulating	Biological Control[d,e]	Pest Control		4		"They eat the pests."; "They eat damaged fruits and help clean up."; "They eat bugs so I like them."	I like this bird because it helps eats pests off my crops	4.53 ± 0.25
	Pollination[d,e]	–		–		–	I like this bird because it helps with pollination	4.22 ± 0.23
Disservice	Crop damage[e]	Harms crops		21		"They damage the mature bananas."; "They eat the crops."; "They damage crops, they eat the best peppers and tomatoes. When we sow, they also destroy."	I don't like this bird because it harms my crops	3.85 ± 0.26
	Invasive plant spread[e]	Disperses invasive plants		2		"They disperse introduced seeds."; "They are harmful. They eat fruits like guayaba and blackberry and seeds like bag weeds and they go dropping them in their poo."	I don't like this bird because it disperses invasive plants (i.e. blackberry)	4.30 ± 0.26
	Annoyance	–		–		–	I find this bird annoying because it forms large groups with other birds	3.25 ± 0.25
		–		–		–	I don't like this bird because it is too noisy	2.25 ± 0.15
	Disease transmission[d,e]	Brings disease to farm		1		"They have brought disease to my chickens recently, the pox."	I don't like this bird because it brings disease to my farm	3.22 ± 0.24
Yellow Warbler								
Cultural	Activity[a,b]; Place[a,b]	Behaviour		14		"The build nests in the coffee."; "(I like them) for their agility when they fly. They can flap their wings and remain immobile."; "They like bathing in water, sometimes I put water out so they can bathe."; "They are curious and like playing in the car mirror."	I like this bird for their company	6.52 ± 0.07
		Brings joy		7		"It makes me happy to see them."; "They bring joy to life."; "I like seeing them."		
		Companion		7		"They're next to me all day."; "Sometimes it sits on my shoulder or my head."; "They are the farmers co-worker (buddy). They are very close."		
	Aesthetic[a,c]	Pretty	Colour not mentioned	32	18	"They are pretty."	I like this bird because it is pretty or/and has nice songs	6.54 ± 0.07
			Colour mentioned		14	"(I like this bird) for their colour."; "Their colour is attractive."; "Their colour, they are the only ones that have colour."		
		Nice songs		8		"It sings nicely."; "They sing the most."		
		Small bird		1		"It is a small bird."	–	–

(Continues)

TABLE 4 (Continued)

Ecosystem service	Theme (type of benefit, value, or negative effect)	Code (trait)	*n*	Examples of responses	Likert statement	Likert score (mean + SE)
	Cultural heritage[a]; Identity[a,c]	Native/Endemic	4	"They are from here."; "They are native."	I like this bird because it is endemic/native to Galapagos	6.56 ± 0.07
	Existence value[a]	Existence value	1	"It is part of nature."	I like this bird because it is part of the ecosystem	6.48 ± 0.07
		Concern for species decline	4	"Before there were many more *marias*."; "What a shame for the taxi drivers who kill them."; "Sometimes the cars kill them."	–	–
	Tourism[d]	Value to tourism	1	"Tourism depends on this and we have to take care (of them)."	This bird is good because it can help for ecotourism	6.50 ± 0.07
	Bequest[a,c]	–	–	–	This bird is important for future generations	6.63 ± 0.07
	Spiritual value[a]	–	–	–	This bird provides spiritual value	5.79 ± 0.17
Regulating	Biological Control[d,e]	Pest control	11	"They help us control pests."; "They help maintain the population of insects in equilibrium."; "They look for worms/caterpillars."	I like this bird because it helps eats pests off my crops	5.87 ± 0.17
	Pollination[d,e]	–	–	–	I like this bird because it helps with pollination	4.98 ± 0.22
Disservice	Crop damage[e]	Harmless	21	"They don't cause damage."; "They are not bothersome."; "They are inoffensive."	I don't like this bird because it harms my crops	1.83 ± 0.05
	Invasive plant spread[e]	–	–	–	I don't like this bird because it disperses invasive plants (i.e. blackberry)	2.73 ± 0.18
	Annoyance	–	–	–	I find this bird annoying because it forms large groups with other birds	2.06 ± 0.11
		–	–	–	I don't like this bird because it is too noisy	1.86 ± 0.08
	Disease transmission[d,e]	–	–	–	I don't like this bird because it brings disease to my farm	2.04 ± 0.10
Little Vermillion Flycatcher						
Cultural	Activity[a,b]; Place[a,b]	Behaviour	13	"They are agile."; "They are calm."; "It is restless and flies a lot."; "It forms a heart in the air when it wants to impress its mate."	–	–

TABLE 4 (Continued)

Ecosystem service	Theme (type of benefit, value, or negative effect)	Code (trait)		*n*		Examples of responses	Likert statement	Likert score (mean + SE)
	Aesthetic[a,c]	Pretty	Colour not mentioned	47	8	"It is pretty"; "They are rather pretty."; "It is beautiful."	I like this bird because it is pretty or/and has nice songs	6.64 ± 0.07
			Colour mentioned		39	"The colours, they are extraordinary because many animals have sad colours, but their colour is incredible."; "The colour is attractive."; "For their colour."		
		Nice songs		6		"They had a nice song."; "They sing nice."; "It has a nice call when it flies."		
		Small bird		1		"It calls my attention because it is small."	–	–
	Cultural heritage[a]; Identity[a,c]	Native/Endemic		5		"They are endemic."; "They are native."	I like this bird because it is endemic/native to Galapagos	6.62 ± 0.08
		Unique		6		"It is unique."	–	–
		Reminder of home		5		"In Loja it is called *putilla*."; "My son has good memories of this bird because when he was young he went to a farm on Santa Rosa called *Pájaro brujo* and saw them there."; "I have a hat with a nice design of a *brujo* that my children made for me."	–	–
		Symbol for conservation		1		"It has become a symbol and garnered a lot of attention because it is almost gone and is endemic and they are trying to recuperate it."	–	–
		Relationship with tortoises		1		"It used to stand on tortoises back."	–	–
	Existence value[a]	Part of nature		2		"They are birds from the field."; "They decorate the landscape with their colours."	I like this bird because it is part of the ecosystem	6.58 ± 0.08
		Concern for species decline		7		"It is about to disappear."; "It is in danger of extinction."; "I wish there were more."	–	–
	Tourism[d]	Value to tourism		2		"It attracts tourism."; "They are attractive for tourism."	This bird is good because it can help for ecotourism	6.58 ± 0.10
	Bequest[a,c]	–		–		–	This bird is important for future generations	6.68 ± 0.06
	Spiritual value[a]	–		–		–	This bird provides spiritual value	5.88 ± 0.20

TABLE 4 (Continued)

Ecosystem service	Theme (type of benefit, value, or negative effect)	Code (trait)	*n*	Examples of responses	Likert statement	Likert score (mean + SE)
Disservice	Crop damage[e]	Harmless	7	"It does not cause damage."; "They don't harm crops."	–	–
	Invasive plant spread[e]	–	–	–	I don't like this bird because it disperses invasive plants (i.e. blackberry)	2.34 ± 0.17
		–	–	–	I don't like this bird because it is too noisy	0.76 ± 0.09

Note: Codes were identified through the open-ended follow-up question 'Why?' to 'How much do you like or dislike this bird?' and placed into themes drawn from previous literature on types of benefits, values, or negative effects. Codes are matched to Likert Statement responses through the theme. '—' under Likert statement indicates there was no Likert Statement asked that matches the code identified through the open-ended responses. Conversely, '—' under code means that the topic of the Likert statement did not come up through the open-ended question. Green colours represent cultural ecosystem services, yellow represents regulating ecosystem services, and purple colours represent disservices. We used a seven-point Likert-scale where: 1-strongly disagree, 2-disagree, 3-somewhat disagree, 4-neutral, 5-somewhat agree, 6-agree, 7-strongly agree. Darker shades of the colour indicate this thematic code matched a Likert-scale item and lighter shades of colour indicate there was no association between a thematic code and Likert-scale item. Light grey codes must be interpreted with caution since they are grouped into what is considered an anecdotal theme ($n < 10\%$ of respondents; Teff-Seker et al., 2022), but are shown when they correspond to a Likert statement. A complete table including anecdotal themes can be found in supplemental materials (Table S18).

[a]Chan, Guerry, et al. (2012) and Chan, Satterfield, and Goldstein (2012) (Supporting Information S1).

[b]Belaire et al. (2015).

[c]Echeverri et al. (2019).

[d]Millennium Ecosystem Assessment (2005).

[e]Díaz et al. (2018: Supporting Information S1).

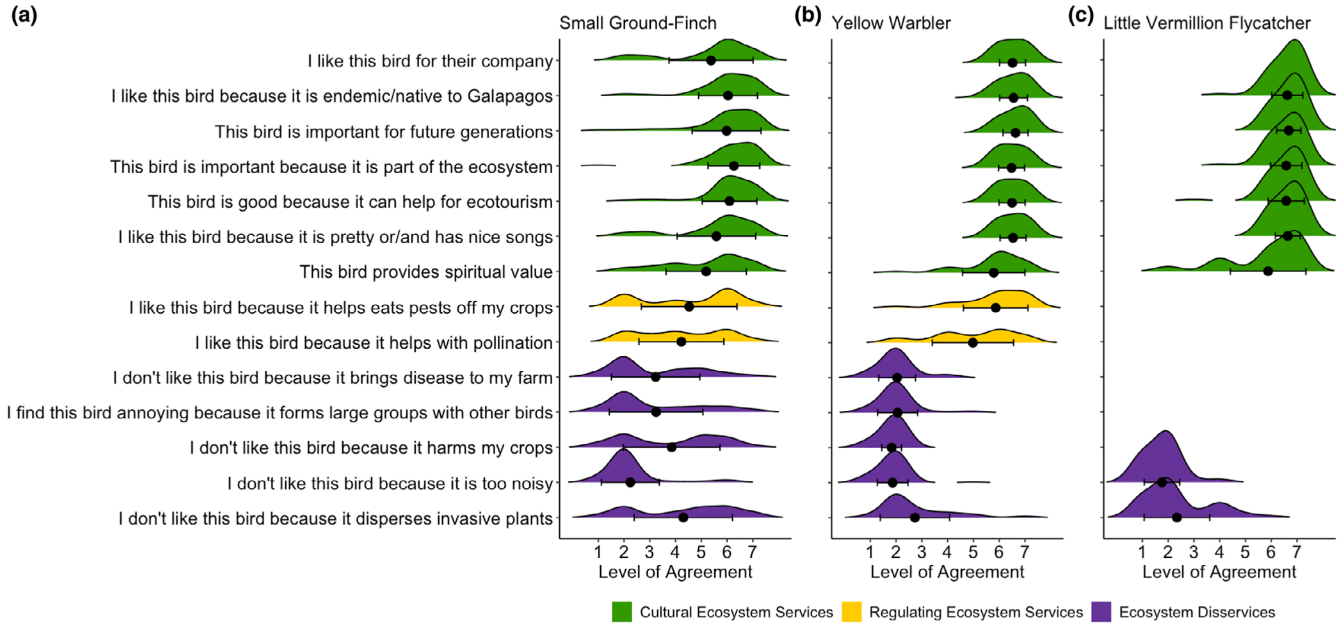

FIGURE 4 Distributions of species-specific Likert-scale responses to questions about cultural ecosystem services (green, first 7 lines), regulating ecosystem services (yellow, lines 8–9), and ecosystem disservices (purple, last 5 lines) for the (a) Small Ground-Finch, (b) Yellow Warbler, and (c) Little Vermillion Flycatcher. Each dot represents the mean value of the level of agreement and the lines represent standard deviations from the mean. The level of agreement was assessed on a seven-point Likert scale: 1-strongly disagree, 2-disagree, 3-somewhat disagree, 4-neutral, 5-somewhat agree, 6-agree, 7-strongly agree.

interviewees) already existed on their property and were intentionally not cut down. Only 35% of interviewees had planted native trees on their property, either independently (8.8%) or as part of a project (22.2%). We also examined trends between farmers' willingness to plant trees and demographic factors (See Section S5; Figure S7).

3.2.2 | What motivated farmers to plant native trees?

When asking farmers, in an open-ended question, what motivates them to want to plant native trees on their property, the most common answer was because these trees can provide shade and/or freshness followed by aesthetic reasons (Figure 5a; Table S10). Most of the motivations mentioned by farmers through open-ended questions were instrumental in nature ($\chi^2_{(3)}=48.5$, $p<0.001$; Figure 5b; Table S11). Unlike open-ended responses, interviewees, on average, expressed stronger agreement with relational value Likert-scale items ($\bar{x}=6.42$, SE = 0.05) compared to instrumental value Likert-scale items ($\bar{x}=5.95$, SE = 0.097; Table S13; Figure 5b). Furthermore, Likert-scale responses revealed that in comparison with other motivational Likert-scale items, farmers were less motivated to plant native trees on their farms because they had heard from other farmers who had benefited from planting native trees ($\chi^2_{(9)}=98$, $p<0.001$; Figure S8; Table S12).

Finally, we also asked farmers whether they would find it important to receive an incentive to plant native trees, and most interviewees said it was 'very important' ($\chi^2_{(4)}=30.3$, $p<0.001$; Figure S9; Table S14). Farmers reported being interested in incentives such as help with manual labour both to plant the trees and for follow-up care, material assistance, technical assistance, and knowledge mobilization (see Table S15 for more details).

3.2.3 | Why were farmers hesitant to plant native trees?

Based on open-ended questions, farmers' main hesitation to planting native trees on their farms was a lack of space followed by the concern that native trees would interfere with production (Figure 5a). The two main types of interference were (1) too much shade and (2) the concern that *Scalesia* trees are fast-growing but short-lived trees that are reported to drop branches and/or fall over, which can damage crops (Table S16). The majority of hesitations were biophysical in nature compared to personal ($\chi^2_{(1)}=11.5$, $p<0.001$; Figure 5c). The Likert-scale responses reflected open-ended question results as interviewees most agreed with '… I do not have enough land to spare' as well as '… I am worried they will interfere with my production' ($\chi^2_{(11)}=87$, $p<0.001$; Figure S8; Table S17).

3.2.4 | Do these motivators and hesitations predict willingness to plant native trees?

The most significant variable associated with farmers' willingness to plant native trees was farmers' instrumental-biophysical

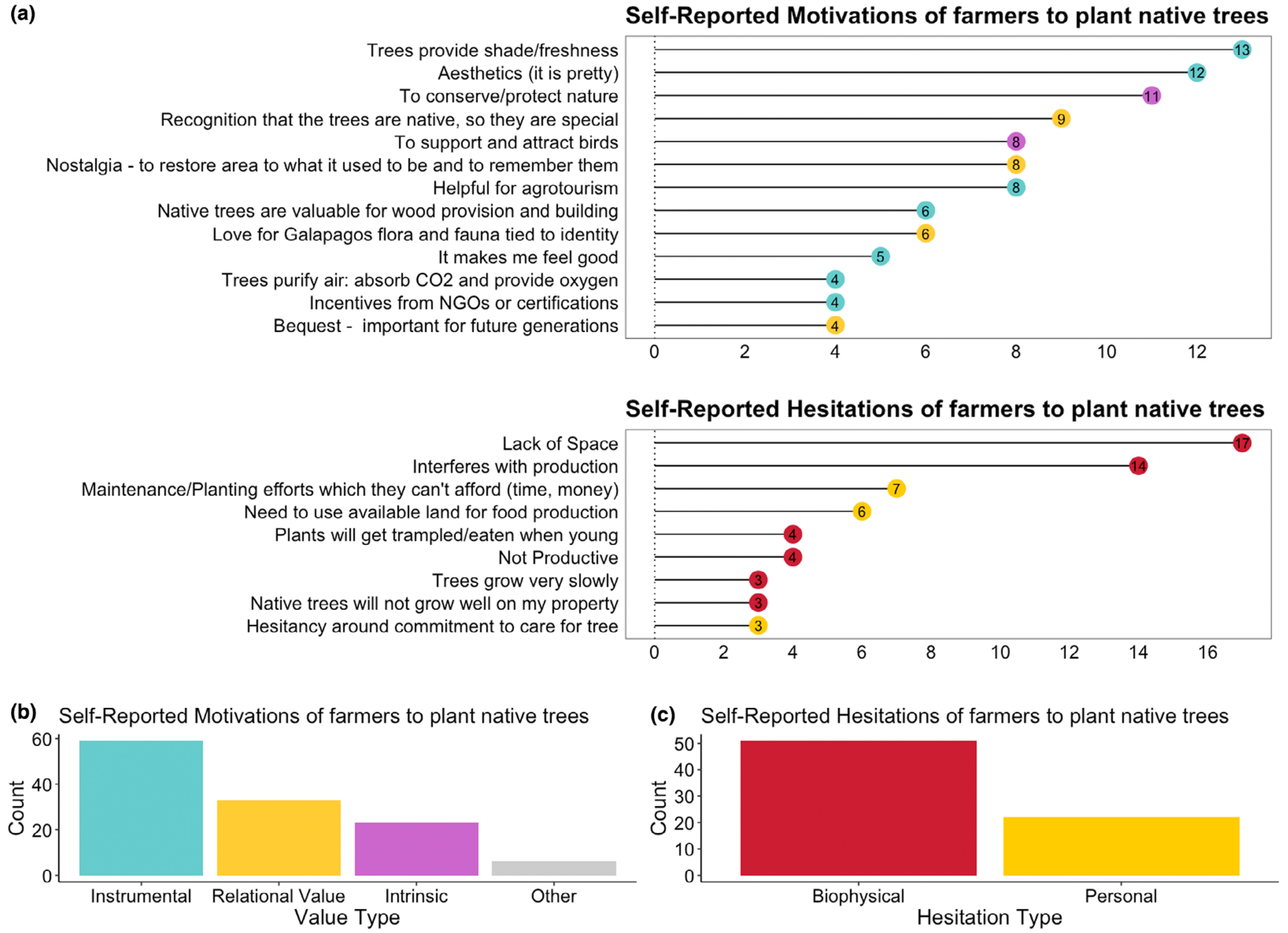

FIGURE 5 Farmers' motivations and hesitations to planting native trees on their farms based on open-ended questions. Panel a shows codes from inductive thematic analysis for open-ended questions about farmers' motivations and hesitations to planting native trees on their farms. Only codes with $n>3$ are shown for motivations and $n>2$ for hesitations (See S10 and S16 for all codes and themes). Panel b depicts all responses from the open-ended question on farmers' motivations grouped into the four possible value types. Panel c depicts all responses from the open-ended question on farmers' hesitations grouped into the two categories.

motivations ($\beta=0.521$, $p=0.005$; Table 3a), although relational value motivations also had a large effect size ($\beta=-0.929$, $p=0.019$; Table 3a). The ability of native trees to provide shade was the predominant Likert-scale item contributing to instrumental-biophysical significance. Other significant variables included instrumental-agrotourism motivations ($\beta=0.260$, $p=0.034$) and biophysical hesitations ($\beta=-0.538$, $p=0.048$; Table 3a).

3.3 | Exploring the relationship between farmers' perspectives on landbirds and willingness to plant native trees

Farmers with more positive perspectives on general landbird services tended to be more willing to plant native trees, whereas those perceiving more disservices were less willing. However, the only statistically significant predictor of tree planting willingness was the presence of native trees. Specifically, farmers who already had native trees on their farms were significantly more likely to plant additional native trees ($\beta=1.727$, $p=0.027$; Table 3b).

4 | DISCUSSION

We studied how small-scale farmers in the Galapagos perceive landbirds and their willingness, motivations, and hesitations to plant native trees on their farms. We found that landbirds provide important CES to farmers and that farmers recognize unique species-specific traits associated with CES. Contrary to expectations, farmers not only recognized traits related to CES but even liked landbirds that provide disservices (i.e. acted as crop pests). Most farmers were willing to plant native trees on their farms, but the extent and willingness were primarily limited due to biophysical limitations such as lack of space. Below, we discuss these results in turn and summarize conservation implications for decision-makers.

4.1 | Farmers' perceptions of Galapagos landbird ecosystem (dis)services

All native landbirds provided several important CES to farmers, even landbirds typically associated with disservice provisioning. Although there was some variation between different landbird species, overall, we found that farmers associated landbirds with traits or characteristics that we interpret as contributing to several CESs. These include identity value (sensu: Chan, Guerry, et al., 2012; Chan, Satterfield, & Goldstein, 2012), as birds were described as bringing joy and being native; activity value (sensu: Chan, Guerry, et al., 2012; Chan, Satterfield, & Goldstein, 2012), as farmers enjoyed their behaviour and companionship; aesthetic value (sensu: Chan, Guerry, et al., 2012; Chan, Satterfield, & Goldstein, 2012) through comments on birds being pretty and singing nice songs; tourism value through birds being helpful for ecotourism (sensu: Belaire et al., 2015); and bequest value (sensu: Chan, Guerry, et al., 2012; Chan, Satterfield, & Goldstein, 2012; Table 4; Figures 2–4). Other studies have found birds provide similar CES to farmers, including identity (~bringing joy, birds being native/endemic), bequest (~important for future generations), and ecotourism value (Echeverri et al., 2019; Zhong et al., 2023). In contrast, Katuwal et al. (2021) found that only about 2% of farmers interviewed in Nepal mentioned any CES provided by birds. This is likely due to the fact that the farmers were not explicitly asked about CES of birds in their study, whereas we specifically asked about CES using Likert-scale questions.

Through open-ended responses, we identified a diverse number of traits associated with CES, which revealed that farmers had detailed perspectives of CES, particularly since they recognized unique species-specific traits associated with CES for different landbird species (Figure S5). For example, farmers liked the Galapagos Mockingbird for its loud sounds and for being fierce, the Woodpecker Finch for its tool-using ability when feeding, and the Dark-billed Cuckoo for its unique song, which farmers believe warns that rains are coming. Although Echeverri et al. (2020) showed that bird traits are significant predictors of some CES, they did not highlight unique species-specific traits that farmers associated with CES. Additionally, they did not explore whether a bird's native or introduced status influenced CES. The Smooth-billed Ani, the only introduced landbird included in our questionnaire, was the most disliked bird, mostly for the perceived damage it does to native landbirds, and the only landbird not reported to provide any CES (Figure S5). It was even more disliked than the endemic Small Ground-Finch, which, despite being known to harm farmers' crops, was still recognized by farmers to provide CES. This finding suggests that the native status of a landbird plays an important role in shaping farmers' perceptions of landbirds, with native species being more likely to be viewed positively and recognized for providing CESs than introduced species. It would be valuable to explore whether a species' native or introduced status influences perceptions of CES in other regions as well.

There was less consensus on the provisioning of regulating services of landbirds on both a general (Figure 2) and species-specific level (Figure 4). This finding was surprising as other studies have shown that farmers generally have the most knowledge of regulating ecosystem services that affect their farming operation (Bernués et al., 2016; Katuwal et al., 2021). However, we found that pest control perceptions had the largest effect on landbird ecosystem service perspectives (Table 2a), highlighting the importance of farmers' perceptions of this service. Nevertheless, we acknowledge the limitations of our study, including the relatively small sample size and the fact that our model only included the perceptions captured by the specific questions we asked; pollination, for example, was not included in general Likert-scale items. One important idea that was mentioned by some farmers is that regulating services, particularly pest control, are only effective if there is a high enough abundance of landbirds to 'make a difference'. Given 55% of interviewees brought up, unprompted, concerns about landbird declines, there might be variation in farmers' perceptions of pest control services depending on: (1) their perceptions of landbird abundances and/or declines; and/or (2) whether they think meaningful services can only be provided if landbird abundance is high enough.

Finally, some farmers did perceive landbirds to be an economic burden and bothersome due to the harming of crops, which was mainly due to finches, particularly the Small Ground-Finch. Farmers did not perceive disservices associated with noise or landbirds being unpleasant to look at (Figures 3 and 4), except for the introduced Smooth-billed Ani (Figure S5). However, perceived losses of agricultural crops from pests often exceed actual losses (Whelan et al., 2016); thus, a study quantifying the actual landbird damages to agricultural crops would provide an interesting complement to this study. In addition, landbird disservice perceptions were associated with crop harm perceptions, as well as Galapagos identity and education, which decreased disservice perceptions (Table 2b). Other studies have also found that education and nature relatedness influenced disservice perceptions (Leong et al., 2020; Zhong et al., 2023), but no previous studies have found identity to be associated with bird disservice perceptions specifically. Leveraging Galapagos identity could be an effective strategy for reducing landbird disservice perceptions, especially given that other studies have found that self-identity influences farmers' willingness to participate in conservation-friendly practices (Cullen et al., 2020; Li et al., 2024). Finally, open-ended responses revealed that the lack of disservices, birds being 'harmless', was important for farmers' liking birds or not (Figure 3). The lack of an ecosystem service or disservice has not been previously considered (Belaire et al., 2015; Echeverri et al., 2019; Zhong et al., 2023) in Likert-scale items and could be an interesting consideration in future studies.

4.2 | Farmers' motivations and hesitations to planting native trees on their farms

Farmers were generally willing to plant native trees on their farms, although typically, only in small quantities. Farmers' willingness to plant native trees was mainly associated with biophysical hesitations (negatively) and instrumental motivations (positively). The main biophysical hesitation was lack of space followed by interference with production (Figure 5). Most lack of willingness to plant native trees came from fruit and vegetable farmers, the owners of the smallest farm type on Galapagos (Geladi et al., 2022). Interference with production encompassed different concerns, the most common being that *Scalesia* is a short-lived tree, which breaks and then falls on crops (Table S16). This concern was particularly true for coffee farmers who had planted *Scalesia* between coffee plants. However, farmers also raised concerns about planting Guayabillo, another endemic tree, as it is very slow growing. These findings suggest that considering different native species on a farm-to-farm or land-use basis, and working with farmers' individual risk tolerance is important.

Farmers' main motivations for planting native trees were instrumental, such as for shade/freshness or aesthetics, followed by 'to conserve/protect nature' (Figure 5; Table S10). These findings were similar to Morgan and Ries (2022) who found beauty and helping the environment were two important motivations for farmers in tree-planting. Galapagos farmers also strongly agreed with most relational value Likert-scale items related to motivations for planting native trees, highlighting the importance of the specific Galapagos context. However, relational value Likert-scale items were unexpectedly negatively associated with farmers' willingness to plant native trees (Table 3). This negative association could be spurious due to the small sample size and will need a confirmatory test with a larger sample in future studies. Farmers who identified as Galapagueño/a most strongly agreed with planting native trees (Figure S7), similar to the influence of Galapagos identity on perceptions of landbird disservices. Contrary to expectations and other studies, social norms (here, measured by a Likert-scale item assessing motivation to plant native trees because of social influence; Figure S8) were the least motivating factor for farmers to plant trees (Buyinza et al., 2020; Villamayor-Tomas et al., 2019). This unexpected result might stem from too few farmers having adopted or benefited from planting native trees for social norms to have a perceived influence. Alternatively, it could stem from conflicting perceptions of success among farmers who have participated in tree-planting projects, particularly with *Scalesia*, or perhaps there is simply not enough communication between farmers for social norms to be an influence. Finally, similar to other studies (Oduro et al., 2018), we found that incentives were important for farmers as a motivation to plant native trees (Tables S8 and S15).

4.3 | Do farmers' perceptions of landbird ecosystem services and disservices influence their willingness to plant native trees?

Although there was a trend towards farmers with higher perceptions of landbird services and lower perceptions of landbird disservices being more willing to plant native trees, the strongest correlate of willingness to plant native trees was whether farmers already had native trees on their farms (Table 3b). This finding aligns with the literature on behaviour spillover. A meta-analysis by Maki et al. (2019) found that individuals who engaged with one pro-environmental behaviour were likely to adopt a second one, especially if both behaviours were similar. In our case, most farmers who already had native trees on their farms had them because they made the decision not to cut them down, and only a few had actually planted native trees on their property. The extent to which people have conserved native trees may thus be helpful in identifying farmer typologies for likelihood of adoption, a helpful prioritization approach when starting conservation projects (Upadhaya et al., 2021). We found that the amount of trees (native or non-native) already present on a farm was not correlated to willingness to plant native trees (Table 3b), in opposition to other studies (Conway, 2016; Villamayor-Tomas et al., 2019). These results may reinforce the importance of focussing on native, rather than non-native trees, as a correlate for willingness to plant native trees. Alternatively, the size of the farm may be a bigger limiting factor compared to the number of trees already on the farm.

4.4 | Key takeaways and recommendations for conservation of landbirds in Galapagos

Farmers are the people who live and care for the (agricultural) land and hold strong ties to both the land and its other inhabitants. Working with and understanding their perceptions, motivations, and hesitations provides an important voice for them in decision-making spaces, which is crucial to building successful conservation strategies (Ahnström et al., 2009; Maas et al., 2021). Based on the findings of our study, we outline recommendations for conservation actions and future studies relevant to the conservation of Galapagos landbirds.

First, we suggest that native tree-planting conservation efforts should have an adaptive approach and be flexible with the native tree species they wish to plant. Farmers had varying perspectives depending on the native tree species, in addition to the islands' microclimates influencing where certain species of trees can grow. Thus, we suggest follow-up research investigating the potential of each native tree species for varying uses on farms (i.e. *Scalesia* vs. Guayabillo for live fences).

Second, based on our findings that the main barrier to planting native trees was a lack of space, we suggest targeting larger farms, where farmers might be less hesitant about lack of space, as a starting point for tree planting. Collaborating with larger farms also directly benefits landbirds, since it is easier to plant trees in patches or rows, and because many large farms are pasture farms which have been identified as priority conservation areas due to lack of diverse landbird communities (Geladi et al., 2021). Additionally, trees can provide a wide variety of benefits to livestock (Broom et al., 2013). Many farmers also expressed interest in agrotourism, where there is much potential to work with farmers to plant more trees to attract more landbirds. Other CES studies on birds reported bird-watching as a CES (Echeverri et al., 2019; Michel et al., 2020; Zhong et al., 2023), which we believe is an untapped potential in Galapagos, where the famous Darwin's finches could attract tourists.

Third, the provision of support and incentives would be important components of a tree planting initiative; for example, a fund to support tree planting initiatives on farms, which would ideally be led by a local organization. However, we did find that farmers were quite willing to help plant trees on other people's farms due to a lack of space on their own farms, provided community events were organized. Therefore, we suggest organizing community events among farmers to plant trees, especially for the initial labour-intensive tree planting phase. These community events could also contribute to increased social norms around tree planting, which we did not find in our study.

Fourth, CES are linked to people's motivations to care for nature (Chan, Guerry, et al., 2012); thus, we recommend using our findings to incorporate the CES of landbirds that farmers value into conservation messaging in Galapagos to promote conservation behaviours, such as planting native trees. Specific bird traits, such as song, colour, and behaviour influenced how farmers perceived CES and can be used to build emotionally resonant, locally grounded messages that connect tree planting to the presence of beloved bird species. Importantly, such messaging should be paired with strategies that address key barriers to tree planting identified above, while also promoting factors such as community planting events, agrotourism benefits, and flexibility in tree species selection. Conservation messages informed by our findings may also be relevant for urbanites of Santa Cruz (Echeverri et al., 2019). We do not recommend the traditional knowledge-deficit approach (Kidd et al., 2019) since farmers are aware of the declining landbird population and knowledge was not the main limiting factor in planting native trees. Finally, species-specific traits or CES with the highest levels of agreement could also be used to support other conservation goals, such as reducing vehicle speeds on highways, a known source of landbird mortality (García-Carrasco et al., 2020).

Fifth, we suggest further studies exploring what 'being Galapagueño/a' means. Self-identity is a factor that has been found to drive pro-environmental behaviours (Carfora et al., 2017; Li et al., 2024), and indeed, we found that farmers who identified more as Galapagueño perceived less landbird disservices and were more willing to plant native trees on their farms. In fact, a local program, '*acuerdos de conservación*', is working to incentivize farmers to plant endemic trees with the goal of cultivating farmers' 'Galapagos identity' (Conservation International, 2016). If we want to understand how to leverage Galapagos identity and the downstream effect for conservation, this construct requires further exploration.

Finally, farmers' varying perceptions on regulating services by landbirds have highlighted the need to better understand and define the role landbirds play on farms in Galapagos. Whilst some studies have shown that Galapagos landbirds play an important role in pollination (Hervías-Parejo & Traveset, 2018; Traveset et al., 2015), this research has been focused on arid plant species. Furthermore, there are no studies, to our knowledge, investigating the potential role of Galapagos landbirds as pest control agents. We recommend further research on bird-mediated pollination and pest control within the agricultural zone, as well as research that quantifies landbird disservices and explores management strategies to prevent disservices.

5 | CONCLUSION

In conclusion, we found farmers' perceptions of Galapagos landbirds varied with species and were overall more positive than expected. Most farmers were willing to plant native trees on their farms; however, they felt limited by available space. We found much potential on Santa Cruz Island to work with farmers towards landbird conservation, which would be strengthened by incentives and an adaptive approach to restoration and/or tree-planting in agricultural zones.

With the expansion of agriculture globally and the associated decline in bird populations (Stanton et al., 2018), it is critical that we consider the socio-cultural dimensions of biodiversity (Pascual et al., 2017) and adopt interdisciplinary approaches that incorporate social and behavioural sciences to develop effective conservation strategies (Balmford et al., 2021; Bennett et al., 2017). Our study adds empirical evidence to the relatively limited body of research on human valuation of birds through CESs. Our study also addresses calls to integrate behavioural science with conservation science by examining stakeholders' motivations and hesitations in the context of a specific conservation goal, planting native trees.

AUTHOR CONTRIBUTIONS

Ilke Geladi conceived the study and designed the methodology with input from Claire Kremen, Jiaying Zhao, Birgit Fessl and Sandra Garcia; Ilke Geladi collected the data with logistical help from Sandra Garcia; Ilke Geladi analysed the data advised by Jiaying Zhao and Claire Kremen; Ilke Geladi led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known conflicts of interest that could influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data has been made publicly available on Borealis: https://doi.org/10.5683/SP3/OAX0NP.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Section S1. Study area.

Section S2. Survey design.

Section S3. Model selection approach.

Section S4. Bird services and disservices as perceived by farmers.

Section S5. Farmers' willingness to plant trees and demographic factors.

Table S1. Friedman test results and Pairwise Wilcoxon signed rank test between Likert statements for general views held by farmers on bird ecosystem services and disservices.

Table S2. Pairwise chi square test results to test if farmers' responses to crop damage questions by birds significantly differed from one another (Figure S2).

Table S3. Pairwise chi square test results to test if responses to farmers' perspectives on pest control services provided by birds significantly differed from one another (Figure S3).

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Table S9. Pairwise chi square test results for the question: 'On how much of your land, in terms of percentage, would you be willing to plant more trees (of any kind)?' asked to people who were willing to plant trees on their farms.

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Table S15. Codebook and frequency of codes for the types of incentives farmers would be interested in.

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Table S17. Friedman test and pairwise Wilcoxon signed rank test results for hesitations to planting native trees on their farm Likert statements.

Table S18. Complete table including anecdotal themes of species-specific landbird ecosystem (dis)service perspectives of farmers.

Figure S1. Map of Agricultural Zone of Galapagos and different neighbourhoods from Geladi et al. (2022).

Figure S2. Bar graphs depicting farmers' responses to various questions on crop damages by birds.

Figure S3. Bar graphs depicting farmers' responses to various questions on pest control services provided by birds.

Figure S4. Bar graphs showing distribution of in response to question asking farmers 'How much do you like or dislike this bird?' for all non-focal bird species.

Figure S5. Inductive thematic analysis for open-ended questions for all non-focal species asking farmers 'Why do you like or dislike this bird?'

Figure S6. Farmers' willingness to plant native trees on their farms.

Figure S7. Stacked bar charts examining farmers' willingness to plant native trees on their farms based on different demographic grouping factors: (A) farm type: (B) Galapagos Identity; (C) Education level; and (D) amount of trees based on farm type.

Figure S8. Distribution of farmers' motivations and hesitations to planting native trees: Likert responses according to value-type.

Figure S9. Bar chart showing importance of receiving incentives to farmers for planting native trees on their farms.

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